

HONORS PHYSICS: ALPINE RESCUE RELAY

Names: _____

Period: 0

Date: _____

UNIT 2: SCALAR PATH DISTANCE VS. VECTOR DISPLACEMENT

• RACE DYNAMICS

EMERGENCY DISPATCH: ALPINE MOUNTAIN RELIEF OPERATION

A distress beacon is activated by an injured mountaineer located at coordinates **(800 m East, 600 m North)** from Base Camp (designated as origin **0, 0**). Two emergency response units depart Base Camp at the exact same moment (**t = 0 s**):

- **All-Terrain Ground Rover Alpha:** Must follow a treacherous, winding switchback canyon road to avoid ravines. The total odometer road distance is **d_{road} = 1,800 meters**. The rover maintains a maximum steady speed of **v_{rover} = 15.0 m/s** along the road.
- **Autonomous Sky-Crane Drone Beta:** Flies straight as an arrow directly from Base Camp over the forest canopy to the climber's coordinates at a cruise speed of **v_{drone} = 20.0 m/s**. However, due to high altitude rotor heating, the drone must land on a midway ridge for a mandatory **35-second thermal cooldown pause** (at the 500 m mark).

PART 1: 2D VECTOR DISPLACEMENT VS. 1D SCALAR DISTANCE

PYTHAGOREAN THEOREM

Step 1A: Calculate the straight-line displacement magnitude (**Δr**) for Drone Beta:

Components: Δx = 800 m (East), Δy = 600 m (North)
 $\Delta r = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{[(800)^2 + (600)^2]} = \sqrt{[640,000 + 360,000]} = \sqrt{[1,000,000]} = \text{_____ meters}$

Step 1B: Compare the path distances of both vehicles:

- Rover Alpha Total Path Distance (scalar): **d = 1,800 meters**
 - Drone Beta Total Path Distance (vector magnitude): **Δr = 1,000 meters**
- Why is the rover's path distance 800 meters greater than its net displacement?
- _____

PART 2: STEP-BY-STEP TRAVEL TIME CALCULATIONS

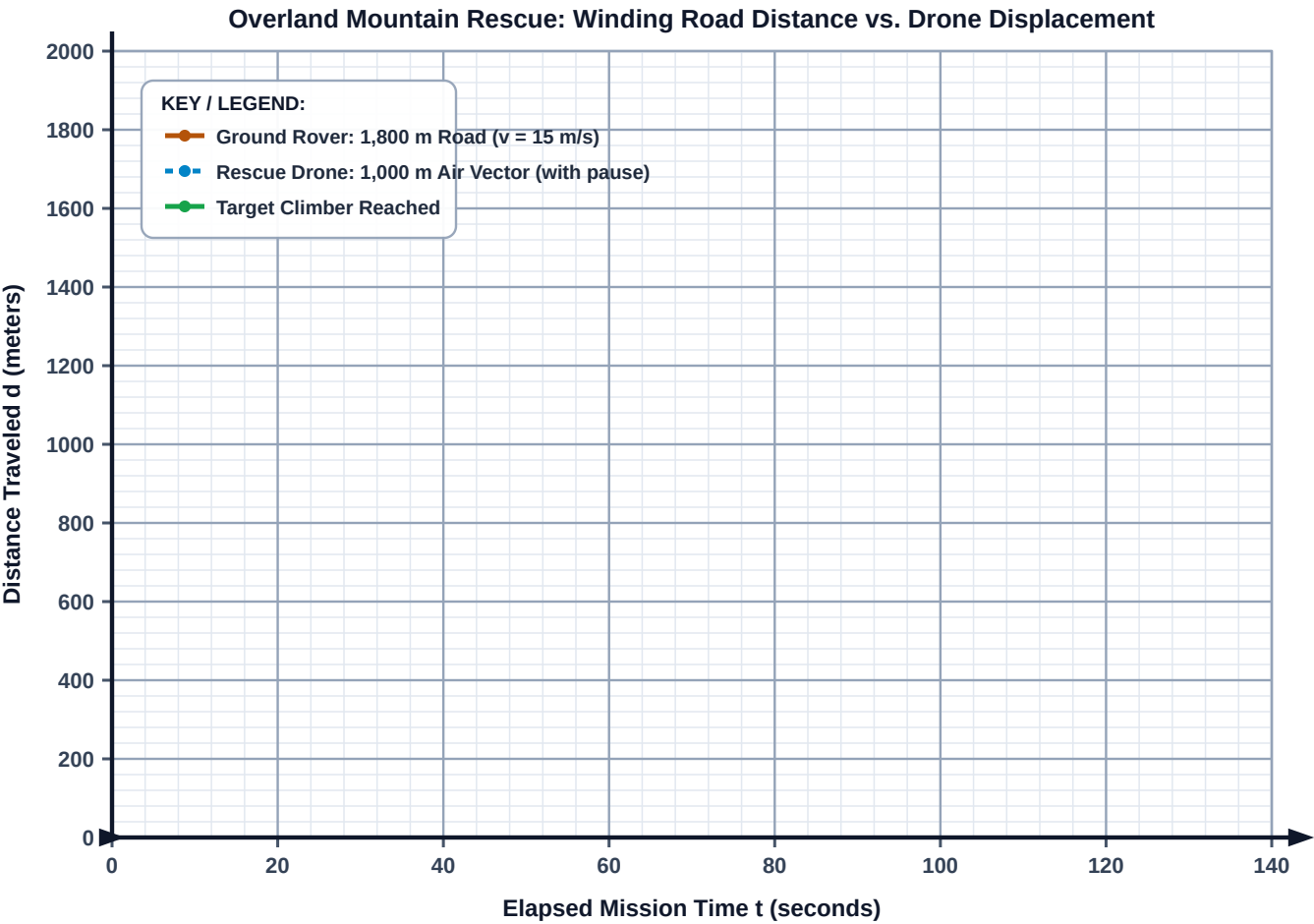
KINEMATIC DERIVATION

Step 2A: Calculate Rover Alpha's total transit time to reach the climber:

$t_{\text{rover}} = (\text{Total Road Distance}) / (\text{Rover Speed}) = 1,800 \text{ m} / (15.0 \text{ m/s}) = \text{_____ seconds} \text{ (_____ min)}$

Step 2B: Calculate Drone Beta's multi-stage timeline (Leg 1 → Cooldown → Leg 2):

- Leg 1 Flight (0 to 500 m): $t_1 = 500 \text{ m} / 20.0 \text{ m/s} = \textbf{25.0 seconds}$ (Clock time: 0 to 25 s)
 - Thermal Battery Cooldown: Duration = **35.0 seconds** (Clock time: 25 s to 60 s)
 - Leg 2 Flight (500 to 1,000 m): $t_2 = 500 \text{ m} / 20.0 \text{ m/s} = \textbf{25.0 seconds}$ (Clock time: 60 s to 85 s)
- Total Drone Arrival Time:** $t_{\text{drone}} = 25 \text{ s} + 35 \text{ s} + 25 \text{ s} = \text{_____ seconds}$



PART 4: MISSION DEBRIEF & MARGIN OF VICTORY

TEAM SYNTHESIS

1. Margin of Victory:

Which vehicle delivers medical aid to the climber first? Calculate the exact margin of victory in seconds:

2. Interpreting the Flat Line on a Distance-Time Graph:

On your graph, describe Drone Beta's trajectory between $t = 25 \text{ s}$ and $t = 60 \text{ s}$. What does the zero slope represent physically, and how does distance accumulated change during this interval?

3. The Critical Breakdown Threshold (Honors Extension):

Suppose high mountain winds increase the drone's thermal cooldown time. What is the **maximum possible cooldown duration** the drone could spend on the ridge and still arrive at the climber before the rover? **Show your mathematical proof:**